

Llama-3.3-70B VIS Benchmark

without aiDAPTIV vs with aiDAPTIV (Cold / Warm)

This report compares serving performance with and without aiDAPTIV under two aiDAPTIV states: Cold (first run) and Warm (steady-state average).

1) Benchmark scope

	Model	Llama-3.3-70B-Instruct
	NP levels	16 and 32
	Input tokens	1K, 2K, 4K, 8K, 16K
	Output tokens	1K and 2K

2) Comparison modes

	without aiDAPTIV	3-run baseline average (Run1 / Run2 / Run3)
	with aiDAPTIV (Cold)	Run1 (first run after enabling aiDAPTIV)
	with aiDAPTIV (Warm)	Average of Run2 and Run3 (steady state)

3) Metrics in this report

	TTFT (ms)	Time from request start to the first token.
	TPOT (ms/token)	Average time to generate each token after the first token.
	Throughput (Aggregate tok/s)	Total server output capacity across all users.
	Throughput (Normalized tok/s/user)	Aggregate throughput divided by NP, representing per-user serving capacity.

4) Reading guide

- ✓ TTFT and TPOT: lower is better.
- ✓ Aggregate throughput: total server output capacity.
- ✓ Normalized throughput: aggregate throughput divided by NP, representing per-user serving capacity.
- ✓ Warm-state gains become more visible as input length and concurrency increase.

 This report focuses on NP16 and NP32 serving behavior under short-to-long context workloads.

TTFT

Time to first token (ms). Lower is better. Comparison modes: without aiDAPTIV, Cold, and Warm.

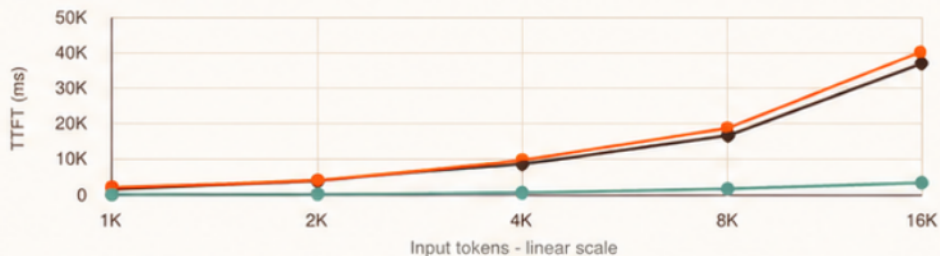
Warm = average of Run2 and Run3; Cold = Run1.

— without aiDAPTIV — Cold (Run1) — Warm (Run2/Run3 avg)

FORMULA $TTFT = t(\text{first token}) - t(\text{request start})$

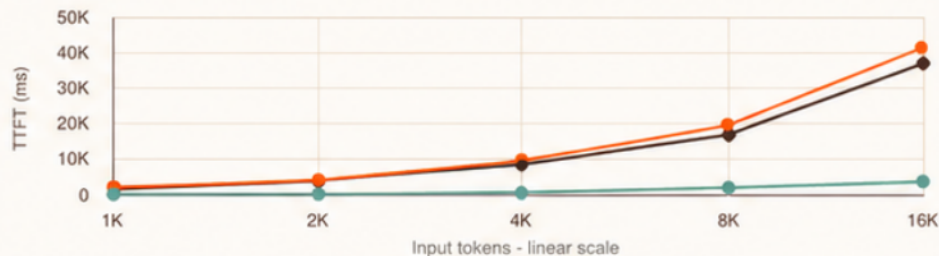
Reduction (%) = $(\text{Without} - \text{Warm}) / \text{Without} \times 100$

NP16 — Output 1K



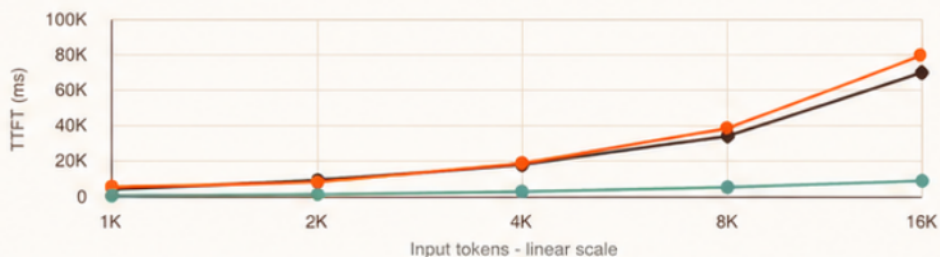
Input	Without (ms)	Cold (ms)	Warm (ms)	Improvement vs Without (lower is better)
1K	2,254	2,728	157	+93.0% lower
2K	4,932	3,292	319	+93.5% lower
4K	9,373	10,412	749	+92.0% lower
8K	18,780	20,777	1,739	+90.7% lower
16K	37,406	41,238	3,312	+91.1% lower

NP16 — Output 2K



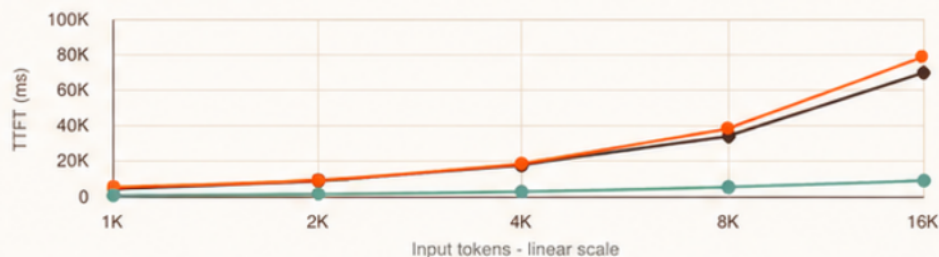
Input	Without (ms)	Cold (ms)	Warm (ms)	Improvement vs Without (lower is better)
1K	2,554	2,938	153	+94.0% lower
2K	4,857	5,392	303	+93.8% lower
4K	8,995	10,034	433	+95.2% lower
8K	18,667	20,750	1,612	+91.4% lower
16K	37,429	41,162	3,276	+91.3% lower

NP32 — Output 1K



Input	Without (ms)	Cold (ms)	Warm (ms)	Improvement vs Without (lower is better)
1K	4,668	5,210	293	+93.7% lower
2K	8,825	5,585	604	+93.2% lower
4K	17,331	19,235	1,386	+92.0% lower
8K	35,064	39,011	3,622	+89.7% lower
16K	73,009	79,655	7,632	+89.6% lower

NP32 — Output 2K



Input	Without (ms)	Cold (ms)	Warm (ms)	Improvement vs Without (lower is better)
1K	4,604	5,157	274	+94.1% lower
2K	8,904	9,901	732	+91.8% lower
4K	17,238	19,124	1,259	+92.7% lower
8K	35,305	38,960	3,511	+90.1% lower
16K	72,699	79,712	6,935	+90.5% lower

TPOT

TPOT

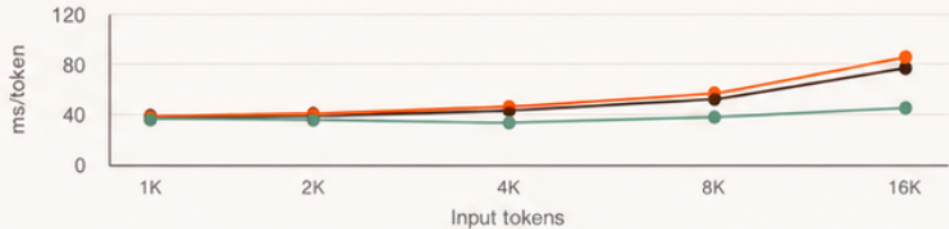
Time per output token (ms/token). Lower is better; smaller values *mean faster* decode.
Warm = average of Run2 and Run3, Cold = Run1.

Without (baseline avg) Cold (Run1) Warm (avg of Run2/Run3)

FORMULA $TPOT = (E2E\ latency - TTFT) / (output\ tokens - 1)$

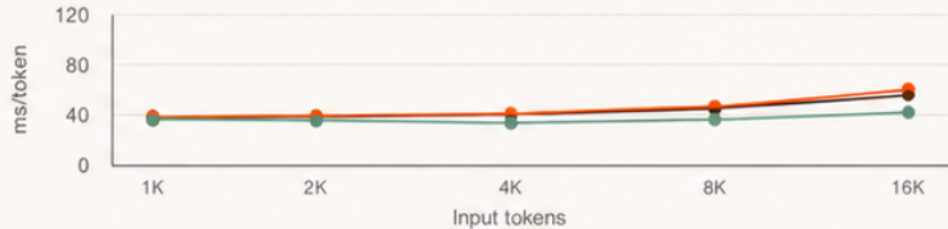
Reduction (%) = $(Without - Warm) / Without \times 100$

NP16 — Output 1K



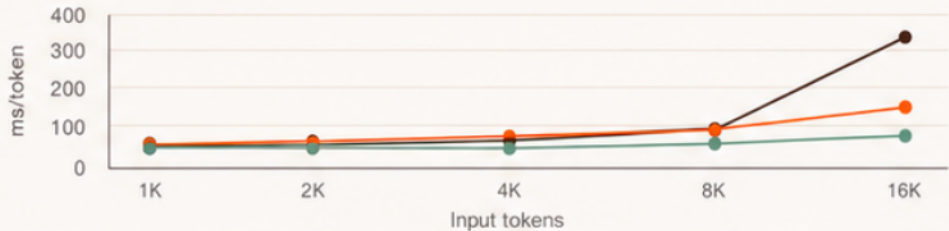
Input	Without (baseline avg)	Cold (Run1)	Warm (avg)	Improvement vs Without (lower is better)
1K	38.74	39.06	38.02	+1.9% lower
2K	41.15	39.92	39.09	+5.0% lower
4K	47.42	48.64	41.92	+11.6% lower
8K	58.18	60.31	45.21	+22.3% lower
16K	87.82	92.34	58.17	+33.8% lower

NP16 — Output 2K



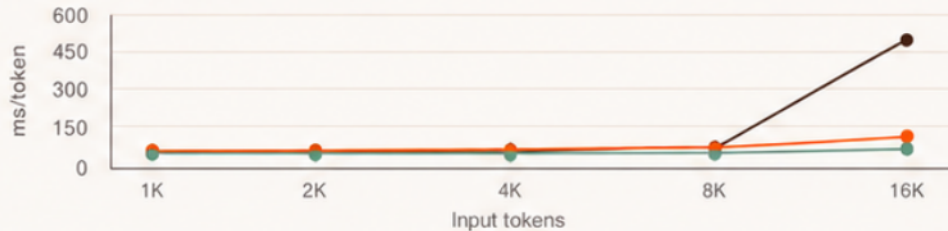
Input	Without (baseline avg)	Cold (Run1)	Warm (avg)	Improvement vs Without (lower is better)
1K	38.43	38.93	38.28	+0.4% lower
2K	42.24	42.72	41.39	+2.0% lower
4K	44.22	45.02	41.91	+5.2% lower
8K	52.47	53.94	46.02	+12.3% lower
16K	67.83	70.52	54.64	+19.4% lower

NP32 — Output 1K



Input	Without (baseline avg)	Cold (Run1)	Warm (avg)	Improvement vs Without (lower is better)
1K	47.88	48.52	45.56	+4.8% lower
2K	53.13	50.45	47.43	+10.7% lower
4K	65.09	67.34	51.92	+20.2% lower
8K	89.47	94.16	60.15	+32.8% lower
16K	332.23	150.38	79.71	+76.0% lower

NP32 — Output 2K



Input	Without (baseline avg)	Cold (Run1)	Warm (avg)	Improvement vs Without (lower is better)
1K	47.36	47.85	46.36	+2.1% lower
2K	51.03	51.87	48.01	+5.9% lower
4K	58.67	60.28	52.49	+10.5% lower
8K	75.00	77.82	60.75	+19.0% lower
16K	491.80	119.33	82.92	+83.1% lower

Throughput (Aggregate)

Aggregate output throughput (tok/s). Higher is better; this is total server serving capacity.

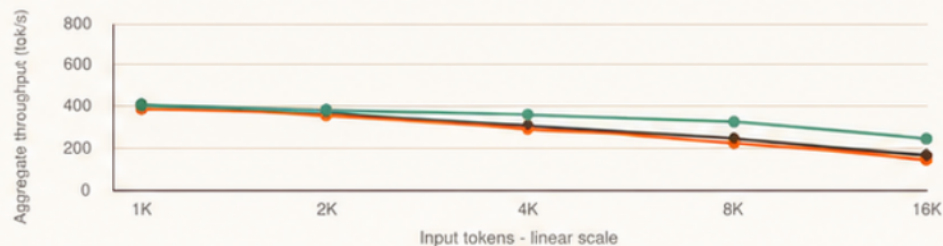
Warm = average of Run2 and Run3, Cold = Run1.

— without aiDAPTIV — Cold (Run1) — Warm (Run2+Run3)

FORMULA Aggregate tok/s = total generated output tokens / benchmark duration

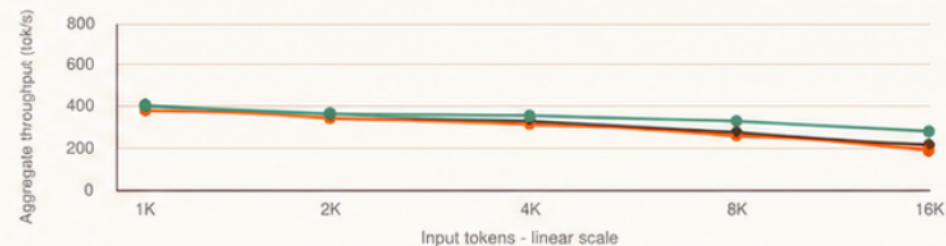
Gain (%) = (Warm / Without - 1) x 100

NP16 — Output 1K



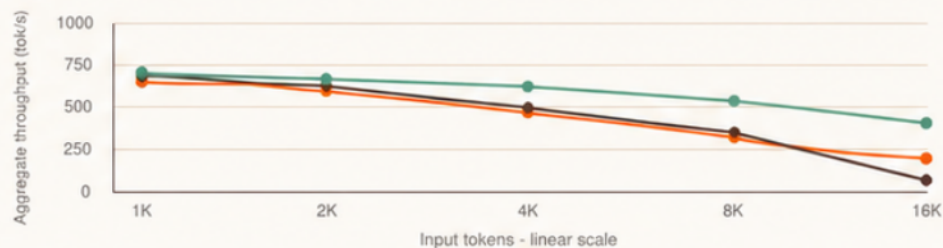
Input	Without (tok/s)	Cold (tok/s)	Warm (tok/s)	Improvement vs Without (higher is better)
1K	412.96	409.60	420.80	+1.9%
2K	388.80	400.80	409.36	+5.3%
4K	337.44	328.96	381.68	+13.1%
8K	274.99	265.28	353.92	+28.7%
16K	182.19	173.28	275.04	+51.0%

NP16 — Output 2K



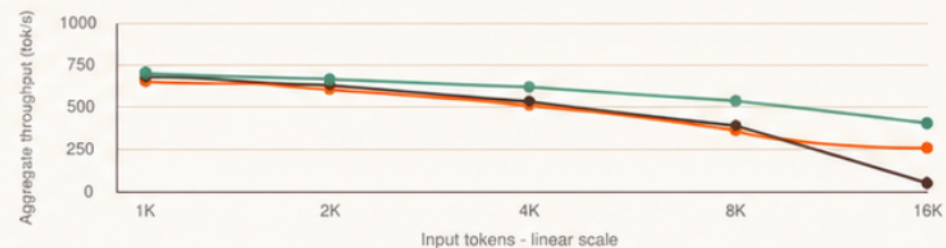
Input	Without (tok/s)	Cold (tok/s)	Warm (tok/s)	Improvement vs Without (higher is better)
1K	416.32	411.04	418.00	+0.4%
2K	378.83	374.56	386.56	+2.0%
4K	361.81	355.36	381.76	+5.5%
8K	304.96	296.64	347.68	+14.0%
16K	235.89	226.88	292.80	+24.1%

NP32 — Output 1K



Input	Without (tok/s)	Cold (tok/s)	Warm (tok/s)	Improvement vs Without (higher is better)
1K	668.37	659.52	702.40	+5.1%
2K	602.35	634.24	674.72	+12.0%
4K	491.63	475.20	616.32	+25.4%
8K	357.65	339.84	532.00	+48.8%
16K	96.32	212.80	401.44	+316.8%

NP32 — Output 2K



Input	Without (tok/s)	Cold (tok/s)	Warm (tok/s)	Improvement vs Without (higher is better)
1K	675.63	668.80	690.24	+2.2%
2K	627.09	616.96	666.56	+6.3%
4K	545.39	530.88	609.60	+11.8%
8K	426.67	411.20	526.72	+23.5%
16K	65.07	268.16	385.92	+493.1%

Throughput (Normalized)

Normalized output throughput (tok/s/user). Higher is better; this is aggregate throughput divided by NP.

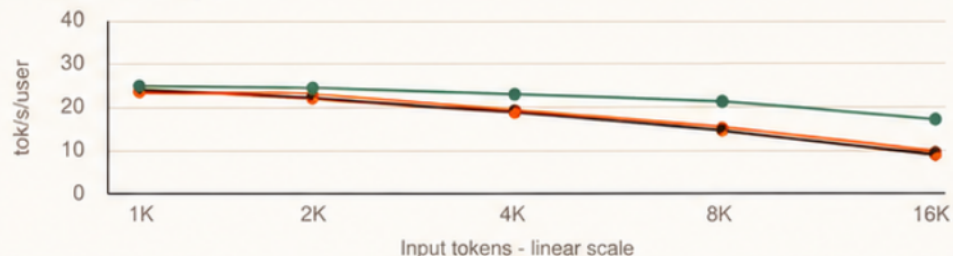
Warm = average of Run2 and Run3; Cold = Run1.

— without aiDAPTIV — Cold (Run1) — Warm (Run2+Run3)

FORMULA Normalized tok/s/user = Aggregate tok/s / NP (concurrent users)

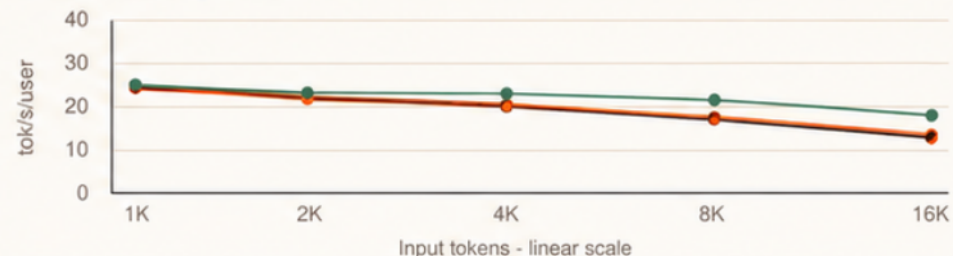
Gain (%) = (Warm / Without - 1) x 100

NP16 — Output 1K



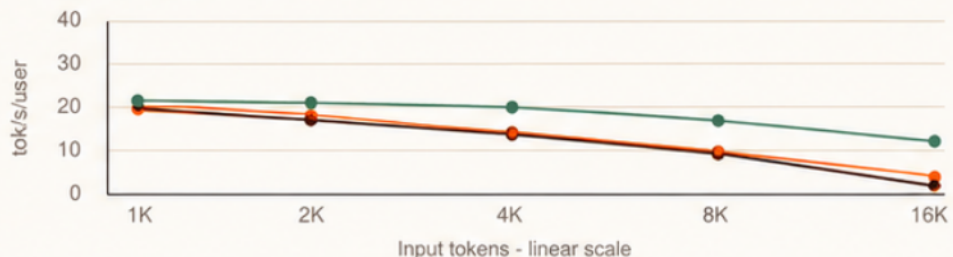
Input	Without (tok/s/user)	Cold (Run1) (tok/s/user)	Warm (Run2+Run3) (tok/s/user)	Improvement vs Without (higher is better)
1K	25.81	25.60	26.30	+1.9%
2K	24.30	25.05	25.58	+5.3%
4K	21.09	20.56	23.86	+13.1%
8K	17.19	16.58	22.12	+28.7%
16K	11.39	10.83	17.19	+51.0%

NP16 — Output 2K



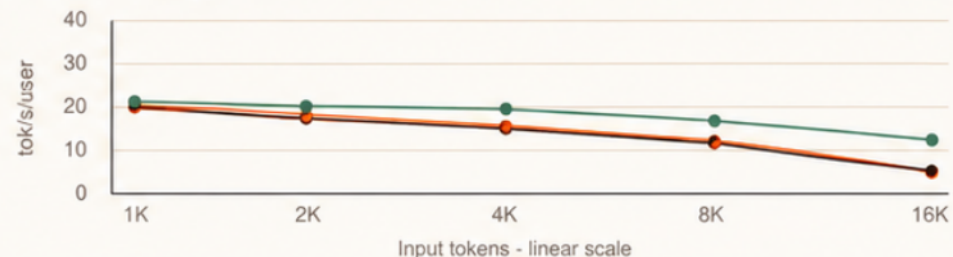
Input	Without (tok/s/user)	Cold (Run1) (tok/s/user)	Warm (Run2+Run3) (tok/s/user)	Improvement vs Without (higher is better)
1K	26.02	25.69	26.12	+0.4%
2K	23.68	23.41	24.16	+2.0%
4K	22.61	22.21	23.86	+5.5%
8K	19.06	18.54	21.73	+14.0%
16K	14.74	14.18	18.30	+24.1%

NP32 — Output 1K



Input	Without (tok/s/user)	Cold (Run1) (tok/s/user)	Warm (Run2+Run3) (tok/s/user)	Improvement vs Without (higher is better)
1K	20.89	20.61	21.95	+5.1%
2K	18.82	19.82	21.08	+12.0%
4K	15.36	14.85	19.26	+25.4%
8K	11.18	10.62	16.62	+48.8%
16K	3.01	6.65	12.54	+316.8%

NP32 — Output 2K



Input	Without (tok/s/user)	Cold (Run1) (tok/s/user)	Warm (Run2+Run3) (tok/s/user)	Improvement vs Without (higher is better)
1K	21.11	20.90	21.57	+2.2%
2K	19.60	19.28	20.83	+6.3%
4K	17.04	16.59	19.05	+11.8%
8K	13.33	12.85	16.46	+23.5%
16K	2.03	8.38	12.06	+493.1%

aiDAPTIV impact on latency, decode efficiency, and throughput

Llama-3.3-70B VIS benchmark, NP16 and NP32; without aiDAPTIV vs Cold vs Warm.

TTFT

Time to first token (ms)

Warm aiDAPTIV vs without aiDAPTIV

NP16

Output 1K

90.7%–93.5% lower

Output 2K

91.3%–95.2% lower

NP32

Output 1K

89.6%–93.7% lower

Output 2K

90.1%–94.1% lower



Warm aiDAPTIV consistently reduced TTFT by ~89.6% to 95.2% across all settings.

TPOT

Time per output token (ms/token)

Warm aiDAPTIV vs without aiDAPTIV

Key highlight (NP32, long context)

16K input | Output 1K

332.23 → 79.71

-76.0%

16K input | Output 2K

491.80 → 82.92

-83.1%



Decode efficiency improved most strongly at long context. Up to -83.1% lower TPOT with Warm aiDAPTIV.

Aggregate throughput

Total server output capacity (tok/s)

Warm aiDAPTIV vs without aiDAPTIV

Key highlight (NP32, long context)

16K input | Output 1K

96.32 → 401.44

+316.8%

16K input | Output 2K

65.07 → 385.92

+493.1%



Gains were modest at short input but grew strongly at longer input and higher NP.

Normalized throughput (per-user capacity)

Aggregate tok/s per user (tok/s/user)

Warm aiDAPTIV vs without aiDAPTIV

NP16

16K input | Output 1K

11.39

17.19

+51.0%

16K input | Output 2K

14.74

18.30

+24.1%

NP32

16K input | Output 1K

3.01

12.54

+316.8%

16K input | Output 2K

2.03

12.06

+493.1%



Per-user serving capacity mirrors aggregate gains. Largest gains at long context with NP32.

Key interpretation

- ✓ Cold alone is not consistently better than without aiDAPTIV; the main performance benefit appears after warm-up.
- ✓ Gains are small at 1K input, become clearer from 4K upward.
- ✓ Strongest impact at 8K–16K, especially with NP32.
- ✓ Warm aiDAPTIV dramatically cuts TTFT, improves decode efficiency, and greatly increases throughput and per-user capacity under high concurrency.
- ✓ Use Warm aiDAPTIV in production to unlock full performance at scale.



Metric guide: TTFT / TPOT: lower is better. Throughput: higher is better.